Homework # 5

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Problem set number: 5

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Question 1:

Known:

Radius: $r = 0.001$ m

Length: $L = 0.05$

Ambient temp: $T_\infty = 25^\circ C$

Base temp: $T_b = 125^\circ C$

Heat transfer coeff: $h = 10$ W/m²K

Thermal conductivity: $k = 200$ W/mK

Assumptions:

one-dimensional conduction in the fin

constant properties

steady state

uniform h

insulated tip

Find:

a) Temp at $x = L/2, L$

b) Fin heat transfer rate q_f, fin effectiveness ε_f, fin efficiency η_f

c) Fin efficiency

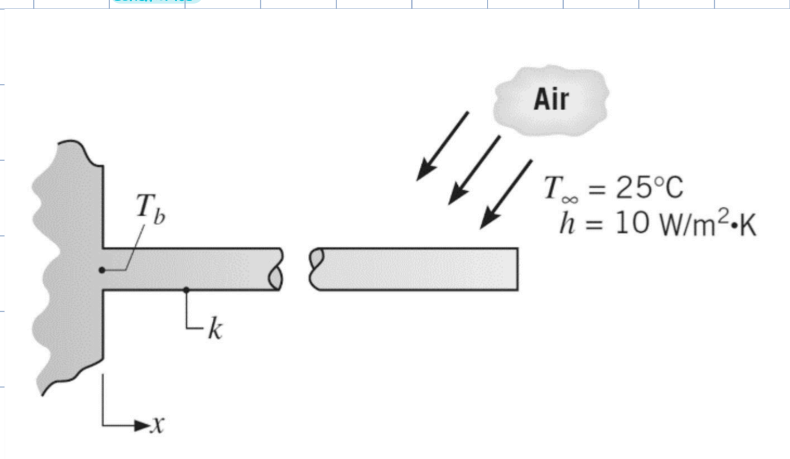
d) Thermal resistance of one pin

e) η_o, R_o, Heat through unfinned plate, heat through fins

f) Thermal resistance network, heater temp, Temp reduction with heat sink

Analysis:

Schematics:



a)

$$A_c = \pi r^2 = \pi (0.001)^2 = 3.142 \times 10^{-6} \, m^2$$

$$P = 2\pi r = 0.006283 \, m$$

$$\theta_b = T_b - T_\infty = 100^\circ C$$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{10(0.006283)}{200(3.142 \times 10^{-6})}} = 10 \, m^{-1} \rightarrow mL = 0.5$$

$$\frac{\theta(x)}{\theta_b} = \frac{\cosh[m(L-x)]}{\cosh(mL)}$$

At $x = L/2$:

$$x = 0.025 \, m$$

$$m(L-x) = 10(0.05 - 0.025) = 0.25$$

$$\frac{\theta(L/2)}{\theta_b} = \frac{\cosh(0.25)}{\cosh(0.5)} = 0.9147$$

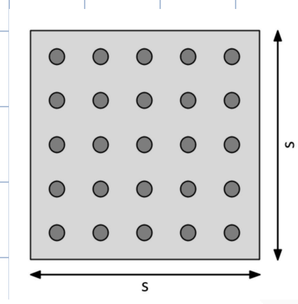
$$\theta(L/2) = 91.47$$

$$T(L/2) = 25 + 91.47 = 116.47^\circ C$$

At $x = L$:

$$\frac{\theta(L)}{\theta_b} = \frac{1}{\cosh(0.5)} = 0.8868 \rightarrow \theta(L) = 88.68$$

$$T(L) = 25 + 88.68 = 113.68^\circ C$$



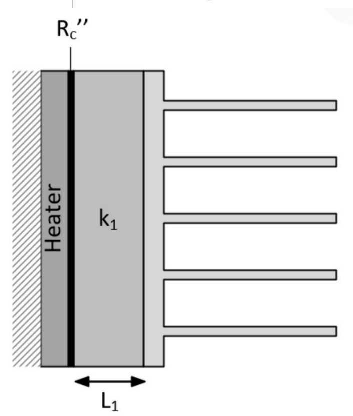
b)

$$q_f = mkA_c\theta_b \tanh(mL) = (10)(200)(3.142 \times 10^{-6})(100)\tanh(0.5) = 0.2904 \, W$$

$$\varepsilon_f = \frac{q_f}{hA_c\theta_b} = \frac{0.2904}{(10)(3.142 \times 10^{-6})(100)} = 92.4$$

$$A_f = PL = (0.006283)(0.05) = 3.142 \times 10^{-4}$$

$$\eta_f = \frac{q_f}{hA_f\theta_b} = \frac{0.2904}{(10)(3.142 \times 10^{-4})(100)} = 0.924$$



c

$$L_c = L + \frac{A_c}{P} = 0.05 + \frac{0.001}{2} = 0.0505\,m$$

The efficiency from table 3.5 is $\eta = 0.923$ and in part b its $\eta_f = 0.924$. The difference is very small (about 0.1%)

showing that the corrected length approximation in Table 3.5 gives an excellent estimate of fin efficiency

$$mL_c = 10(0.0505) = 0.505$$

$$\eta_f = \frac{\tanh(mL_c)}{mL_c} = \frac{\tanh(0.505)}{0.505} = 0.923$$

d

$$R_f = \frac{T_b - T_\infty}{q_f} = \frac{100}{0.2904} = 344.4\ K/W$$

e

$$N = 25,\ A_t = (0.03)^2 = 9\times10^{-4}$$

$$A_b = A_t - NA_c \rightarrow A_b = 9\times10^{-4} - 25(3.142\times10^{-6}) = 8.21\times10^{-4}$$

$$A_0 = A_b + NA_f = 8.21\times10^{-4} + 25(3.142\times10^{-4}) = 8.675\times10^{-3}$$

$$\eta_0 = 1 - \frac{NA_f}{A_0}(1-\eta_f) = 1 - \frac{25(3.142\times10^{-4})}{8.675\times10^{-3}}(1-0.924) = 0.931$$

$$q_0 = \eta_0 h A_0 (T_b - T_\infty) = (0.931)(10)(8.675\times10^{-3})(100) = 8.08\ W$$

$$R_0 = \frac{100}{8.08} = 12.38\ K/W$$

unfinned

$$q = hA_b(T_b - T_\infty) = (10)(8.21\times10^{-4})(100) = 0.821\ W$$

fins

$$q = Nq_f = 25(0.2904) = 7.26\ W$$

f

$$Q = q'' A = (1000)(9\times10^{-4}) = 0.9\ W$$

$$R_c = \frac{10^{-4}}{9\times10^{-4}} = 0.111$$

$$R_{cond} = \frac{0.02}{(10)(9\times10^{-4})} = 2.22$$

$$R_{tot} = R_c + R_{cond} + R_0 = 0.111 + 2.22 + 12.38 = 14.71$$

$$T_h = T_\infty + QR_{tot} = 25 + (0.9)(14.71) = 38.2°C$$

Question 2:

Known:

Wire diameter $D = 0.001\,m$

Oil temp $T_\infty = 20°C$

Current $I = 100\,A$

Resistance per unit length $R'_e = 0.01\,\Omega/m$

Heat transfer coefficient $h = 500\ W/m^2K$

Density $\rho = 4000\ kg/m^3$

Heat capacity $c = 500\ J/kgK$

Thermal conductivity $k = 400\ W/mK$

Assumptions:

Long wire

Uniform temp in cross section

Lumped capacitance valid

Constant properties

Find:

a Steady state wire temperature

b Temp expression $T(t)$, time to reach within 5°C of steady state

Analysis:

a

$q' = I^2 R_e' = 100 \text{ W/m}$

$A' = \pi D$

$100 = (500)(\pi)(0.001)(T_{ss} - 20)$

$T_{ss} - 20 = 63.66 = 83.66^{\circ}C$

b

$L_c = \frac{D}{4} = 2.5 \times 10^{-4}$

$Bi = \frac{hL_c}{k} = 3.1 \times 10^{-4} \ll 0.1$

$\tau = \frac{\rho c D}{4h} = 1s$

$T(t) = T_\infty + (T_{ss} - T_\infty)(1 - e^{-t/\tau}) = 20 + 63.66(1 - e^{-t})$

$63.66e^{-t} = 5 \rightarrow t = \ln\left(\frac{63.66}{5}\right) = 2.54$

Question 3:

One homework assignment that I found extremely valuable was Homework 5, which focused on fin heat transfer and transient heating of a wire. This assignment helped me better understand how fins improve heat dissipation and how concepts such as fin efficiency, fin effectiveness, and thermal resistance are used to evaluate the performance of heat sinks. Comparing the exact fin solution with the corrected length approximation showed how simplified engineering models can still produce highly accurate results.

Another key concept from this assignment was the lumped capacitance method which is used to analyze the transient heating of the wire. This problem reinforced the importance of the Biot number in determining whether the lumped capacitance assumption is valid and showed how transient temperature behavior can be modeled using exponential relationships. Overall, this homework strengthened my understanding of both steady state and transient heat transfer analysis and demonstrated how these methods are used to evaluate practical thermal systems.